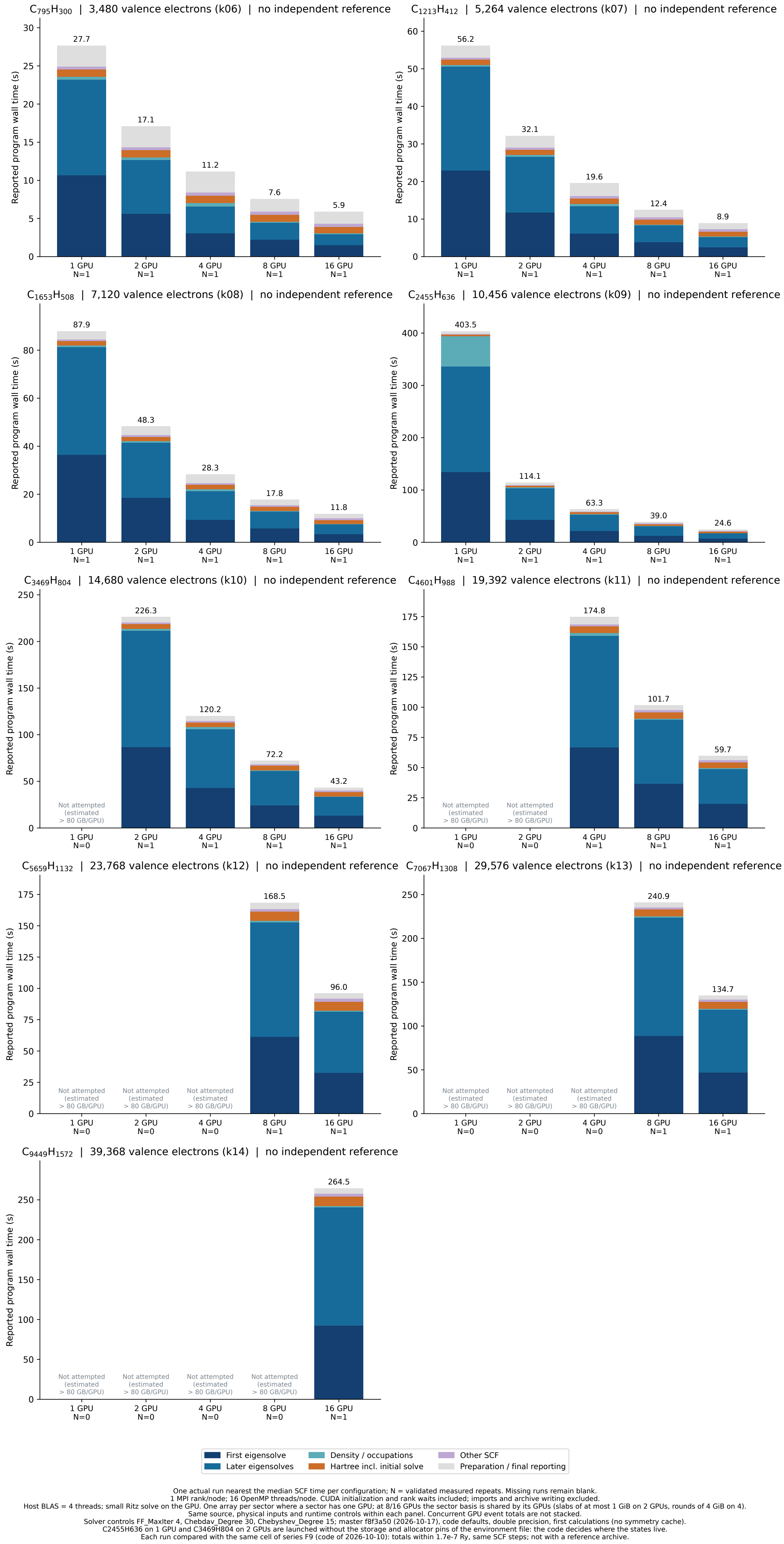
Complete SCF wall-time breakdown
Converged, physics-validated runs | all symmetry sectors | FP64



One actual run nearest the median SCF time per configuration; N = validated measured repeats. Missing runs remain blank.
1 MPI rank/node; 16 OpenMP threads/node. CUDA initialization and rank waits included; imports and archive writing excluded.
Host BLAS = 4 threads; small Ritz solve on the GPU. One array per sector where a sector has one GPU; at 8/16 GPUs the sector basis is shared by its GPUs (slabs of at most 1 GiB on 2 GPUs, rounds of 4 GiB on 4).
Same source, physical inputs and runtime controls within each panel. Concurrent GPU event totals are not stacked.
Solver controls FF_MaxIter 4, Chebdav_Degree 30, Chebyshev_Degree 15; master f8f3a50 (2026-10-17), code defaults, double precision, first calculations (no symmetry cache).
C2455H636 on 1 GPU and C3469H804 on 2 GPUs are launched without the storage and allocator pins of the environment file: the code decides where the states live.
Each run compared with the same cell of series F9 (code of 2026-10-10): totals within 1.7e-7 Ry, same SCF steps; not with a reference archive.